FROM THE
EDITOR-IN-CHIEF

A lot of attention these days is on the global charts India is climbing and getting close to the top. Here's one peak we would rather not be on: India is now the world's largest plastic polluter, according to the journal *Nature*. How we achieved this dubious distinction requires an explanation. India is only about the world's fourth-highest in terms of the amount of plastic waste it generates; the US, China and the EU bloc generate much more than we do. However, what lets India down is the primitive waste management practices that prevail here compared to other polluters. Studies show only about 12 per cent of our plastic waste gets processed. An astounding 88 per cent remains, piling up at open dumpsites, choking our rivers, or getting incinerated by households and informal waste collectors. That's where we become the top plastic polluter on the planet. According to *Nature*, we produce plastic emissions of 9.3 million tonnes per annum. We burn an estimated 5.8 Mt of that, releasing noxious gases. Approximately 3.5 Mt ends up as loose plastic debris. This is not counting the enormous amounts disposed of in open dumps, literally spreading poison. That's a frightening array of toxins leaching permanently into our land, water, air and food chains.

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For comparison, look at China. Its plastic production may be 4-6 times that of India. Even till six years ago, it was tagged as the world's largest plastic polluter. But its annual plastic emission is now only 2.8 Mt, reflecting substantial progress in the adoption of better recycling techniques and stricter regulatory protocols. By contrast, though India's stated per capita consumption of 13 kg is less than half of the global average of 30 kg, its recycling sector is riddled with stark inefficiencies. Open dumpsites outnumber sanitary landfills by 10:1.

India cannot do without plastics. Modern economies pretty much come gift-wrapped in this versatile, durable and lightweight substance. It is an inalienable part of everyday utilities, from packaging material to mobile phones, computers, cars, ships, planes, plumbing, flooring, all manner of high-end industrial uses, right up to spacecraft. The plastics industry was worth Rs 3.5 lakh crore in FY23, accounting for 1.4 per cent of the GDP. India has over 30,000 plastic processing units, close to 90 per cent of these being small and medium enterprises. Together, they employ 4 million people. It's an industry growing at a healthy clip of 10-15 per cent; India even exported plastics worth $12.5 billion in FY25.

Based on their behaviour when heated, plastics can be divided into two broad categories—thermoplastics, which become soft upon heating, can be melted and reshaped and are therefore recyclable; and thermosets like bakelite which can be moulded only once and are used for electrical insulation, cookware, roofing and auto and industrial components. Both types are non-biodegradable. They stay around forever. But, over time, they fragment into toxic micro- and nano-sized particles that infiltrate the soil, the oceans and the food chain, causing serious health issues in humans and animals. Global studies have shown that microplastics are entering the bloodstream and accumulating in organs such as the brain and heart.

To reduce plastic use, the Indian government in 2022 banned 19 single-use plastic (SUP) items with "low utility and high littering potential", including plastic cutlery, straws, trays and decorative thermocol. Additionally, a thickness criterion was imposed on SUP items like carry-bags—only products over 100 microns were allowed. In an effort to check the amount of plastic ending up as debris or poisonous smoke, the government notified the Guidelines on Extended Producer Responsibility (EPR) for Plastic Packaging the same year, making it compulsory for producers, importers and brand owners (PIBOs) to ensure 100 per cent of life disposal of the plastic generated from their products' packaging.

But, like with everything else in India, good intentions falter at the doorstep of implementation. Studies show India's SUP ban addressed hardly 11 per cent of the products in the category, and that carry-bags that were not compliant with SUP thinness rules were still commonly available in major cities, along with other banned products like thermocol decorations and balloons. A major reason for the lack of enforcement, says Swati Singh Sambyal, senior circular economy expert at Norwegian non-profit GRID-Arendal, is the lack of rigour among state pollution control boards and local governments. This, she explains, is compounded by insufficient capacities of the enforcers and manpower. "Additionally," she points out, "the industry for alternatives to plastic remains relatively small and less specialised, with inadequate efforts to integrate it into the mainstream market." SUP alternatives, including bamboo cutlery, areca palm leaf tableware or paper plates, remain expensive, and inaccessible across income groups. Senior Associate Editor Sonal Khetarpal ploughs through mountains of data, sorting and assembling the points that most urgently call us to attention. This week's cover story will hopefully contribute to a workable roadmap. Perhaps that road can be built of recycled plastic.

(Aroon Purie)